

Topology First-Year Exam, January 2013, Part 1

Work the following problems and show all work. Support all statements to the best of your ability.

1. Show that the set of functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ that are eventually constant is countable.
2. Let X be a connected metric space having more than one point. Can X be countable?
3. Show that every continuous map $f : [0, 1] \rightarrow [0, 1]$ has a fixed point. Is this true for continuous maps
 - (a) $f : (0, 1) \rightarrow (0, 1)$?
 - (b) $f : [0, 1] \rightarrow [0, 1)$?
4. Let A be a proper subset of X and let B be a proper subset of Y . If X and Y are connected, show that $(X \times Y) - (A \times B)$ is connected.
5. Let $f : X \rightarrow X$ be a map of a complete metric space to itself that satisfies the following condition: There exists $\lambda < 1$ such that $d(f(x), f(y)) \leq \lambda d(x, y)$. Prove that f is continuous. Show that f has a fixed point and the fixed point is unique.

Answer the following with complete definitions or statements or short proofs.

6. State the Intermediate Value Theorem.
7. Does there exist a continuous surjective map from the 2-sphere S^2 to the interval $(0, 1)$?
8. Does there exist a continuous injective map from the 2-sphere S^2 to the interval $(0, 1)$?
9. State the Cantor–Schröder–Bernstein Theorem.
10. Is every connected space path connected?
11. What is a basis of a topology? Does the set of all half-open intervals $\{(a, b], [b, c) \mid a < b, c < d\}$ form a basis of a topology on $\mathbb{R}$?
12. State the Extreme Value Theorem.
13. Is $\mathbb{R}^\omega$ connected in the uniform topology?
14. Is the subspace $X = \{(x, y) \in \mathbb{R}^2 \mid y = \frac{1}{x}\}$ compact? locally compact?
15. Describe all connected subsets of $\mathbb{R}$.